TITLE: Phan Tuấn kiệt 2352654 Date: HW3

Activity 1:

$x = (x_0, x_1, x_2) = (1; 1; -1)^T$

$W = (0,1; -0,3; 0,4)^T$

$w^T . x = 0,1.1 - 0,3.1 - 0,4.1 = -0,6$

activation function: $a = \begin{cases} 1 & z \geq 0 \\ 0 & z < 0 \end{cases}$

$z = -0,6 < 0 \Rightarrow a = -1$ equal with target

$\Rightarrow$ no need to perceptron update

Activity 2:

$z = \{-3; -0,5; 0,2\}$

$ReLu(z_1) = 0$ Sigmoid $(z_1) = \frac{1}{1+e^{-(-3)}} \approx 0,0474$

$ReLu(z_2) = 0$

$ReLu(z_3) = 0,2$ Sigmoid $(z_2) = \frac{1}{1+e^{-(-0,5)}} \approx 0,3775$

Sigmoid $(z_3) = \frac{1}{1+e^{-0,2}} \approx 0,5498$

Relu have output range $[0; +\infty]$ while output range of sigmoid is $[0; 1]$

Zero use zero gradients to determine how affected of this weight value to the number of Error. In this example with $z = -3$ and $z = -0,5$ the derivative of ReLu $= 0$ which mean that the value of weight can not be update while derivative of sigmoid function for these two value is $0,0452$ for $z = -3$ and $0,2350$ for $z = -0,5$ which mean that the weigh value can be updated updated.

~~The derivation of sigmoid~~ rank and to approaching round to zero ~~to the it~~ it's

The range of rank $[-1; 1]$ while range of sigmoid $[0; 1]$. The range of rank sometime can help system learning more efficient because it have negative value compare to sigmoid which mean that tanh can reduce more effect to the input because it can reduce the value of z.

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Activity 3:

In hidden layer

$z_h = w_h \cdot x + bias = 0,6 \cdot 1 + 0 = 0,6$

the activation of this hidden layer is sigmoid

sigmoid $(0,6) \approx 0,6457$

~~this~~ then let this output into the input for the output layer

$z_o = v \cdot R_h + bias = -0,7 \cdot 0,6457 + 0 = -0,45199$

sigmoid $(0,45199) \approx$ ~~0,6444~~ 0,3889

$Loss = 0,5(t-o)^2 = 0,5(0-0,3889)^2 \approx 0,0756$

Output layer error signal

$\frac{\partial L}{\partial z_o} = \frac{\partial L}{\partial o} \cdot \frac{\partial o}{\partial z}$

here $\frac{\partial L}{\partial o} = (0,5 \cdot (t-o)^2)' = -(t-o) = o$

$\frac{\partial o}{\partial z_o} = o(1-o)$

$\frac{\partial L}{\partial z_o} = 0,3889^2 \cdot (1-0,3889) \approx 0,0924$

$\Rightarrow \frac{\partial L}{\partial v} = \frac{\partial L}{\partial z_o} \cdot \frac{\partial z_o}{\partial v} = 0,0924 \cdot 0,6457 \approx 0,0597 > 0$

$\Rightarrow$ v will reduce. ~~$v_{new} = v - \eta \frac{\partial L}{\partial v} = 0,7$~~

error signal of hidden layer

$\delta_h = \frac{\partial L}{\partial z_o} \cdot v \cdot \sigma'(R_h) = 0,0924 \cdot (-0,7) \cdot 0,6 \cdot (1-0,6) \approx -0,0155$

$\frac{\partial L}{\partial w} = \delta_h \cdot \frac{\partial z_h}{\partial w} = -0,0155 \cdot 1 = -0,0155 < 0$

$\Rightarrow w_h$ will increase.

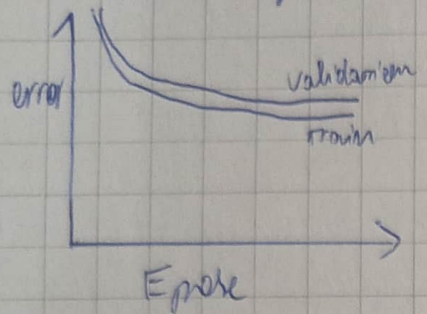
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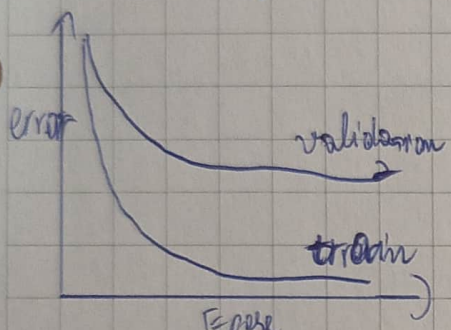
Activity 4:

Under training

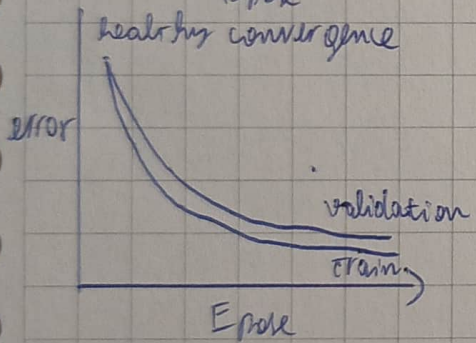


the error can not to reduce even those pass a lot of layer.
To improve it, need add hidden node in the hidden layer to help system can detect more feature of data or add hidden layer.

Over training



the error of train set reduce smoothly but the validation set is stuck.
To improve it reduce the number of node or reduce hidden layer.



Activity 5:

Model A	Model B
maximum hidden layer size = 10	maximum hidden size = 200
maximum hidden width = 500	maximum hidden width = 500
random seed = 31	random seed = 31

Before run code:

because of hidden layer size of model A small than model B so that the time for training model A will small than model B. But the accuracy of training set of model A may smaller than model B because Model B may be become overfitting leading to model B have the smaller validation more than model A.

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After run

~~Mod~~	Model A	Model B
Time	5, 4258 s	4, 1915
Training score	1.0	1.0
Validation score	0, 9355	0, 9326

In the realistic, the time for training model B is small than model A. It may be because Model A need more step for zero gradient leading to Model A need more loop for optimize model, the reason cause this problem may be come from the small number of max hidden_layer_size.
Training score of both models are the same (1.0) and the validation score of Model B is small than model A as predicted before